

STOPPING SILENT ERRORS BEFORE THEY BECOME DECISIONS

Oversight Checkpoints for Hierarchical Multi-Agent AI Systems

Technical work sample and research proposal for the Singapore AI Safety Fellowship

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Background Oxford MPP · Lawyer · Applied AI builder	Domains Colombia · Latin America · Spain	Focus AI evaluations · multi-agent oversight · AI assurance

1. Executive Summary

Large language model agents can produce outputs that are fast, plausible and wrong. In a hierarchical system, a small error at the collection or evaluation stage can be accepted by downstream agents, consolidated into a final recommendation and presented to a human with an appearance of confidence. The most concerning failures are therefore not crashes; they are silent errors that survive every hand-off.

This technical work sample is grounded in a production system that I personally designed and built inside Beta Group. The system coordinates approximately 26 specialised agents across recruitment, tender detection, partner discovery, legal verification and technical-proposal generation in Colombia and Spain. It has processed 50,044 résumé records, deduplicated them into 27,956 unique people, queried an open database containing more than eight million public contracts and supported complex document workflows involving hundreds of pages.

The system already uses safeguards: deterministic data collection where possible, evidence-grounded prompts, structured JSON validation, persistent mission logs, bounded model routing and mandatory human approval for consequential outputs. A follow-up reproducible evaluation on 40 synthetic CVs across gpt-4o-mini and gpt-4o showed that prompt-only fixes were modest and model-dependent, while a deterministic verifier placed in front of the LLM reduced false-positive PASS verdicts to 0 in this sample.

RESEARCH QUESTION
Where should automated checks and human review be placed in a hierarchical multi-agent AI system to stop silent errors before they propagate into consequential decisions, without eliminating the speed and cost advantages that make the system useful?

The proposed research compares no oversight, final-only review, fixed checkpoints and adaptive oversight triggered by risk signals. The intended outputs are a research paper, an open-source evaluation harness, a bilingual benchmark of controlled failures and a working prototype for adaptive oversight.

2. The Operational Problem

Why multi-agent workflows create a distinctive safety challenge

Beta Group is a 14-year public-policy, applied-research and technology consultancy. The firm's work depends on interpreting large volumes of unstructured evidence: résumés, government contracts, tender specifications, legal registries, research documents and client records. Traditional manual review could not scale to the volume or speed required by public procurement.

I began by converting an unstructured internal repository of approximately 6,000 résumés and more than 50,000 supporting documents into structured JSON records. I then developed scraping, classification and search pipelines, and progressively organised them as a hierarchy of specialised agents. The platform now supports high-volume recruitment, partner discovery, tender screening and technical drafting.

Subsystem	Scale / source	Decision-ready output	Safety relevance
Talent evaluation	50,044 résumé records	Eligibility and ranking against tender requirements	False positive may advance an unqualified person
SECOP partner search	8M+ public contracts	Companies with relevant awarded experience	Weak evidence may lead to an unsuitable partner
RUES verification	Public legal registry	Valid bidder registration and renewal status	Extraction error may falsely certify eligibility
Tender analysis	Dense administrative and technical documents	Go/no-go recommendation	Omitted requirement can invalidate a bid
Methodology generation	80-90-page technical proposals	Draft methodology aligned to scoring criteria	Plausible unsupported claims can survive into formal submission

The safety problem is asymmetric. A false negative may exclude a valid opportunity and can often be detected later. A false positive can silently pass an unqualified candidate, unsuitable partner or non-compliant proposal into a consequential workflow. These errors become harder to detect as they move upward, because downstream agents inherit outputs rather than independently reconstructing the original evidence.

DESIGN PRINCIPLE
Use deterministic rules for what can be verified directly; reserve LLM judgement for the genuine long tail; and keep a human in control of consequential commitments.

3. System Architecture and Existing Safeguards

A hierarchy designed for specialisation, observability and bounded autonomy

Capability and autonomy increase by rank; consequential outputs require human approval

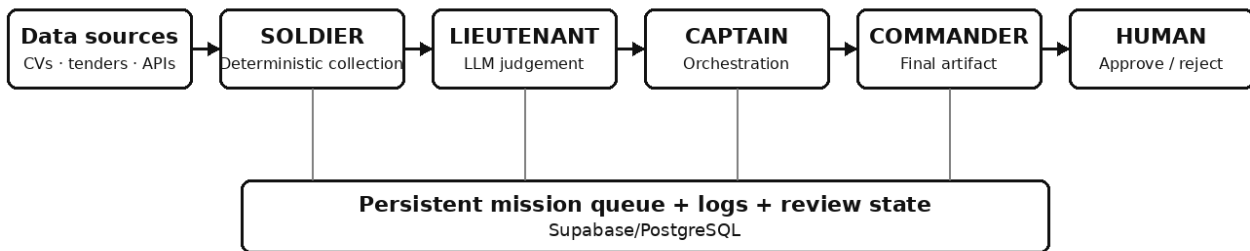


Figure 1. Simplified architecture of the internally built BetaGroup multi-agent system.

The hierarchy encodes responsibility. Soldiers perform deterministic collection and extraction. Lieutenants use LLMs for judgement and structured interpretation. Captains consolidate results and delegate sub-missions. Commanders generate final artefacts. A persistent Supabase/PostgreSQL mission queue decouples the agents, records each hand-off and leaves consequential outputs in a review state for human approval.

- **Rule-first, LLM-last.** A deterministic SNIES taxonomy classifies 99.4% of the résumé corpus without model calls; LLMs handle only the unresolved long tail.
- **Human holds the commit bit.** An agent may propose, score or draft, but consequential outputs remain in review until a person approves them.
- **Observability by construction.** Mission status, progress, heartbeats and outputs are persisted, enabling replay and diagnosis after a failure.
- **Graceful degradation.** Model calls use provider fallback, bounded retries and JSON validation rather than assuming a single model or provider will remain available.
- **Separation of collection and judgement.** Deterministic scraping and extraction are separated from model-based evaluation, making failures easier to localise.

These controls improve reliability, but they do not answer the central research question: which checkpoints materially prevent harmful errors, and which merely add cost and delay? The existing system provides a realistic test bed because the same hierarchy can be run under different oversight regimes while controlled faults are introduced at known locations.

4. Observed Failure Modes and Relevant Assurance Experience

From internal production incidents to a researchable safety problem

Failure mode	Observed example	Potential consequence	Current mitigation
Silent extraction error	A scraper captured a job title instead of a person's name.	Downstream evaluation remained syntactically valid but semantically wrong.	Comparison checks, schema validation and diagnostic logs.
Mis-specified search filter	A wrong filter silently reduced 803 relevant records to 51.	The output looked precise while omitting most of the evidence.	Baseline counts, filter diagnostics and human review of query interpretation.
False-positive LLM judgement	An evaluator passed an unqualified candidate.	A harmful error could survive into a formal tender team.	Evidence-grounded prompt, structured output, deterministic verifier and final expert review.
Malformed or incomplete model output	The model returned prose, invalid JSON or omitted fields.	Agents could fail or persist incomplete evidence.	JSON parser, retries, validation and provider fallback.
Cascading acceptance	A plausible upstream output was treated as fact by later stages.	Confidence increased without independent verification.	Persistent provenance and checkpoints, but no formal evaluation of optimal placement.

My prior AI assurance work provides a complementary institutional perspective. At the Barcelona Supercomputing Center, I directed the ASCENDER ethics and assurance workstream across intelligent mobility, sustainable mobility, precision agriculture and emergency evacuation. The project produced a 23-risk mitigation matrix, a BDD100K dataset audit, an ethical checklist, training and case-specific evaluations aligned with the EU AI Act. The work highlighted risks arising from non-representative data, weak accountability and limited internal capacity to recognise AI harms.

For Colombia's National Planning Department, I led the methodology and research behind a national guide for detecting, preventing and mitigating gender bias across the AI lifecycle. The project translated evidence from interviews, focus groups and literature into lifecycle controls, a tools catalogue and an interactive decision tree. These projects taught me how to convert abstract principles into operational controls; building the internal BetaGroup system showed me that those controls must also be empirically tested.

SAFETY MOTIVATION
The dangerous failure is not that an agent crashes. It is that the system remains fluent, coherent and operational while becoming wrong.

5. Research Question, Scope and Hypotheses

A focused study of error propagation and oversight placement

PLAIN-LANGUAGE QUESTION
If an AI agent makes a small mistake at the beginning of a workflow, where should automated checks or human review intervene so the mistake does not become a major decision?

Formal research question: Which combination of automated verification and human-review checkpoints most effectively prevents silent, high-cost errors from propagating through hierarchical multi-agent systems, while preserving task performance, speed and cost efficiency?

The study focuses on systems in which multiple specialised agents pass structured outputs through a hierarchy. It does not attempt to solve general alignment or model interpretability. Instead, it studies a concrete oversight problem: how to detect and stop harmful errors before they are amplified by delegation, aggregation and final generation.

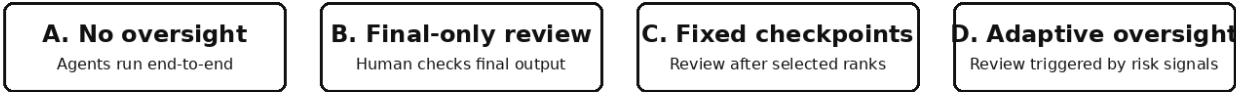
H1	Final-only review will detect fewer upstream errors than checkpointed oversight because the original evidence becomes less visible after consolidation.
H2	Fixed checkpoints will reduce harmful error propagation but impose unnecessary human-review burden on low-risk missions.
H3	Adaptive oversight triggered by missing evidence, contradiction, abnormal result counts or high-risk actions will achieve a better safety-efficiency trade-off than fixed review.
H4	Deterministic verification combined with evidence-grounded LLM evaluation will outperform prompting alone, particularly for explicit legal or eligibility constraints.

The contribution is twofold. First, the research would provide a reproducible benchmark for analysing how errors move through an agent hierarchy. Second, it would produce practical evidence on where human oversight adds the most value, rather than assuming that more review is always safer. This matters for organisations that need AI systems to remain useful under real constraints of time, cost and limited expert attention.

6. Experimental Design

Controlled faults, comparable oversight regimes and measurable outcomes

Experimental comparison of oversight strategies



Controlled faults injected at different levels:

- | | | |
|------------------------|-------------------------|-------------------------|
| • wrong extraction | • missing evidence | • ambiguous CV |
| • mis-specified filter | • contradictory sources | • malicious instruction |

Measured outcomes: harmful error rate · propagation depth · task success · review burden · laten

Figure 2. Proposed comparison of oversight conditions and controlled fault types.

Test bed. A sanitised version of the résumé-evaluation and tender-analysis pipelines will be used. The benchmark will include synthetic English and Spanish documents with labelled ground truth. Synthetic data avoids exposing personal or commercially sensitive information while preserving realistic ambiguity, missing evidence and adversarial structure.

Oversight conditions. The same tasks will be run under four regimes: (A) no review, (B) human review only at the final output, (C) fixed review after selected ranks and (D) adaptive review triggered by risk signals. Each condition will use the same documents, models and task instructions.

Fault injection. Controlled errors will be introduced at the soldier, lieutenant and captain levels. Examples include wrong field extraction, omitted evidence, incorrect filter selection, contradictory documents, fabricated support, malformed structured output and an instruction embedded in an untrusted source. The location and intended effect of each fault will be known in advance.

Models and baselines. At least two commercially available models will be evaluated. Baselines will include naive prompting, evidence-grounded prompting and evidence-grounded prompting combined with deterministic verification. Temperature, tool permissions and retrieval context will be fixed where possible.

Procedure. Each task will be repeated across multiple seeds. Logs will capture the original evidence, intermediate outputs, checkpoint decisions, human interventions and final verdict. The analysis will identify whether an error was detected, where it was detected, how far it propagated and what additional cost or delay the intervention introduced.

CORE COMPARISON
Safety is not measured only by final accuracy. The experiment measures whether the system catches the harmful error early enough, with a review burden that a real organisation could sustain.

7. Preliminary Evidence and Evaluation Plan

A small empirical result that motivates a larger, falsifiable study

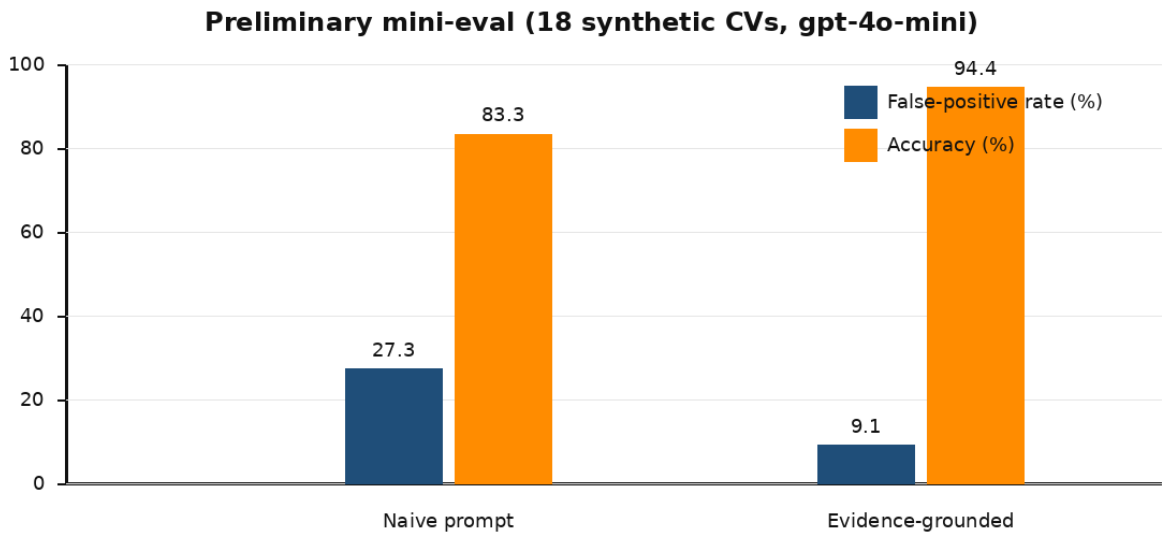


Figure 3. Evidence-grounding reduced harmful false positives in an initial pilot, but a residual rule violation remained.

The current GitHub repository includes a reproducible evaluation of LLM-as-a-judge reliability derived from the internal production system. Forty synthetic CVs were evaluated against strict disqualifying requirements across gpt-4o-mini and gpt-4o. The evaluation compared naive prompting, evidence-grounded prompting and evidence-grounded prompting combined with a deterministic verifier.

Model	Condition	Acc.	FPR	FPR 95% CI	FN	Cost
gpt-4o-mini	Naive	0.925	0.12	[0.04, 0.30]	0	\$0.024
gpt-4o-mini	Grounded	0.925	0.12	[0.04, 0.30]	0	\$0.024
gpt-4o-mini	Grounded + verifier	1.000	0.00	[0.00, 0.13]	0	\$0.024
gpt-4o	Naive	0.950	0.08	[0.02, 0.25]	0	\$0.41
gpt-4o	Grounded	0.975	0.04	[0.01, 0.20]	0	\$0.41
gpt-4o	Grounded + verifier	1.000	0.00	[0.00, 0.13]	0	\$0.41

- The stronger model had a lower baseline error, but prompting alone did not solve the safety problem.
- A deterministic verifier drove false-positive PASS verdicts to 0 for both models in this sample, with no false negatives.
- The cost result matters: gpt-4o-mini plus verifier achieved FPR = 0 at \$0.024, outperforming gpt-4o with prompt-only scaffolding at FPR = 0.04 and \$0.41 - about 17 times the cost.
- A prior 18-CV pilot suggested that evidence grounding alone reduced FPR from 27.3% to 9.1%. Expanding the evaluation changed the conclusion: scaffolding can help, but deterministic verification is the robust separable result when the rule is expressible.

Metric	Operational definition
Harmful error rate	Share of high-cost false positives reaching final output
Propagation depth	Number of agent hand-offs completed before detection
Detection location	Rank or checkpoint where the failure is first identified
Human-review burden	Minutes and number of cases requiring expert attention
Latency and API cost	Incremental time and model cost per oversight regime

The analysis will report point estimates, confidence intervals where sample size permits, model-level comparisons and error categories. Results will be interpreted conservatively. The study is designed to reveal trade-offs and failure boundaries, not to claim universal safety from a small benchmark.

8. Eleven-Week Plan, Deliverables and Expected Contribution

From internal production intuition to rigorous AI safety research

Weeks	Activities
1	Refine threat model; finalise benchmark taxonomy; sanitise test pipelines; preregister hypotheses and metrics.
2-3	Build controlled fault-injection harness; implement baselines; validate ground truth with expert review.
4-5	Run no-oversight and final-only conditions; conduct initial error analysis; revise benchmark difficulty.
6-7	Implement fixed checkpoints, adaptive checkpoints and deterministic verification; run multi-model experiments.
8-9	Analyse safety-efficiency trade-offs; conduct ablations; document failure cases, limitations and cost curves.
10-11	Write paper; release open-source harness and benchmark; produce technical report, demo and implementation guidance.

Planned deliverables

- A research paper on oversight placement and silent error propagation in hierarchical agent systems.
- An open-source evaluation harness with controlled fault injection and replayable mission logs.
- A bilingual English-Spanish benchmark of realistic, labelled failure cases.
- A prototype adaptive-oversight policy using risk signals such as missing evidence, contradiction, abnormal result counts and high-impact actions.
- A practical implementation note for public-sector and enterprise teams deploying agentic workflows.

Expected contribution. The project connects scalable oversight with real deployment conditions. It asks not whether humans should remain involved, but how scarce human attention should be allocated inside a multi-agent workflow. This empirical approach directly aligns with the fellowship's focus on **evals** and **agent alignment**, while also informing the governance of deployed agents. The result would help developers and institutions distinguish review that is merely ceremonial from review that measurably prevents harmful propagation.

Limitations. The initial benchmark will be domain-specific, use synthetic data and evaluate a limited set of models. Human review quality may vary, and adaptive triggers can themselves fail. These limitations will be measured and reported rather than hidden. Generalisation to other domains would require additional replication.

Why I am positioned to execute it. I combine direct system-building experience, public-policy research, law and applied AI assurance. I have led multidisciplinary teams in Latin America and Europe, translated ethical and regulatory principles into operational controls, and personally built the internal production architecture that motivates this research. The fellowship would allow me to convert those operational lessons into rigorous, falsifiable and publicly useful AI safety research.

REPRODUCIBLE EVIDENCE
GitHub repository - jorgegarcia205/betagroup-ai-army Selected supporting work: Barcelona Supercomputing Center ASCENDER AI ethics and assurance workstream (2024-2025); Colombia National Planning Department, Guide to detect, prevent and mitigate gender bias in AI (2025).

Appendix A. Reducing False-Positive PASS Verdicts from an LLM Evaluator

A small empirical study on LLM-as-a-judge reliability

This appendix reproduces the empirical work sample underlying Section 7. It is distilled from the internal production multi-agent LLM system I built and operate, which uses LLMs to evaluate candidates against strict, disqualifying requirements for public tenders. In that setting the dangerous error is the false positive - passing a candidate who should be rejected.

Research question. When an LLM judges whether a candidate meets a requirement, does (a) an evidence-grounded prompt, and (b) a cheap deterministic verifier placed in front of the model, reduce the harmful-direction false-positive error - and does a stronger model need the scaffolding less?

Method

- **Dataset:** 40 synthetic CVs (25 ineligible, 15 eligible), written without giveaways, spanning realistic traps: in-progress, undated and specialization-not-master degrees, wrong-field master, wrong base profession, no master and insufficient experience.
- **Conditions:** (1) naive prompt; (2) evidence-grounded prompt (absence of evidence leads to rejection); (3) grounded plus a rule-based verifier that can only tighten an eligible verdict - rule first, LLM last.
- **Models:** gpt-4o-mini and gpt-4o, temperature 0, five runs per candidate, majority vote.
- **Metrics:** accuracy; false-positive rate with a Wilson 95% confidence interval; false-negative rate; cost; latency.

Results

Model	Condition	Acc.	FPR	FPR 95% CI	FN	Cost
gpt-4o-mini	Naive	0.925	0.12	[0.04, 0.30]	0	\$0.024
gpt-4o-mini	Grounded	0.925	0.12	[0.04, 0.30]	0	\$0.024
gpt-4o-mini	Grounded + verifier	1.000	0.00	[0.00, 0.13]	0	\$0.024
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Findings

- The stronger model has a lower baseline error: gpt-4o had 0.08 FPR compared with gpt-4o-mini at 0.12 FPR.
- Prompt grounding helped only the stronger model, and modestly: gpt-4o improved from 0.08 to 0.04 FPR, while gpt-4o-mini was unchanged.
- A prior 18-CV pilot overstated the effect of prompt grounding. Expanding the evaluation changed the conclusion, which is itself the point of doing evaluations properly.

Appendix A. Findings, Caveat and Takeaway

- A free deterministic verifier drove FPR to 0 for both models, with no false negatives in this sample.
- Cheaper and better than scaling the model: gpt-4o-mini plus verifier (FPR 0, \$0.024) beat gpt-4o with prompt-only scaffolding (FPR 0.04, \$0.41 - about 17 times the cost).
- Wide confidence intervals remain because there were 25 ineligible cases. The point differences between prompts should not be over-read; the robust, separable result is the verifier reaching 0 in this dataset.

Honest caveat and takeaway

The verifier is perfect here because every disqualifier in this dataset is expressible as a rule - by construction. In a real corpus the hard cases are exactly those that are not cleanly rule-expressible, which is why the production system keeps a rule-based layer for the approximately 99% it can decide and reserves the LLM for the residual.

The generalizable lesson is not to assume an LLM judge is reliable. Measure the error that actually hurts, prefer a deterministic guard wherever the decision is expressible as one, and reserve the model and evaluation budget for the fraction rules genuinely cannot decide.

This matters especially for resource-constrained organisations and Global South contexts: a smaller model plus a targeted verifier can be safer and cheaper than simply escalating to a larger model. The lesson is therefore technical and institutional: better AI safety can sometimes come from careful system design, not just from more expensive models.

REPRODUCIBLE CODE, DATASET AND FULL WRITE-UP

github.com/jorgegarcia205/betagroup-ai-army/tree/main/evals

End of technical work sample.